After MID

20/04/2026

- Bandwidth → (i) Analog → range
  (ii) Digital → amount
  - → Baseband BW : Before modulation
    Ethernet
  - → Passband BW : (orginal signal + carrier)
    frequency range
  - → Broadband BW
  - → BW efficiency : $\eta = \frac{R}{B}$ bps/Hz
    (Spectral eff.)
    - R → Information rate
    - B → BW
  - → How many bits can be assigned into 1 Hz

- Power — 2nd Moment

$$E\{x^2\} = \int_{-\infty}^{+\infty} x^2 pdf(x)dx$$

(x is replaced by amplitude)

- Power efficiency → how effectively transmitted power is used
  - → Power Eff. = $\frac{E_b}{N_o}$
    - E_b → Energy per bit
    - N_o → Noise power spectral density
      (thermal Noise pick) → W_o
  - → $E_b = \frac{P}{R}$
    - P → Total energy
    - R → Bit Rate

CamScanner

$$P.E. = \frac{E_b}{N_0} = \frac{P}{R} \cdot \frac{1}{N_0} \quad \ldots (1)$$

Thermal Noise : To calculate N_0, we look here

Modeled as white Noise

$N_0 = k \times T$ → Absolute temperature (K)

↓ Boltzmon constant (JK^{-1})

$$B.E. = \frac{R}{B} \quad \ldots (11)$$

if R increases, PE ↓, BE ↑

(trade-off)

22-04-2026

* $\frac{E_b}{N_0}$ → Efficiency cost calculate

** cost ↓ → Power efficiency ↑ → B.E ↑

P.E ∝ B.E

Power Efficiency vs Bandwidth Efficiency

$$\Rightarrow SNR = \frac{P_{signal}}{P_{noise}} = \frac{P}{N} \quad \text{(linear)}$$

$$SNR_{dB} = 10 \log_{10} \frac{P_{signal}}{P_{noise}} \quad \text{(loganithmic)}$$

High SNR → Strong signal

Low SNR → Weak Signal

⇒ Channel capacity :

$$C = B\log_2(1+SNR)$$

→ maximum number of bits we can transmit in a second.

→ Speed Limit

B = Analog BW, C = Capacity (bps)

→ capacity of Gaussian channel

Gausian Channel : Affected by additive white gaussian noise. (AWGN)

power of spectrum will be same

In ideal cases, we can replace C with bit rate R.

Derivation — (with explanation)

$$C = B\log_2(1+SNR) \quad \ldots (1)$$

In ideal cases;

$$R = B\log_2(1+SNR) \quad \ldots (II)$$

$$\eta = \frac{R}{B} = \log_2(1+SNR) \quad \ldots (III)$$

; η = Bandwith efficiency

In linear scale, $SNR = \frac{P}{N}$ ... (iv)

Noise Bandwidth, $B = \frac{N}{N_0}$ ... (v)
(Noise freq. range)

; $N_0 \rightarrow$ Noise Power spectral density (Watts/Hz)

; $N \rightarrow$ Total / Avg. power of noise (Watt)

Q. Provided BW will be same for signal and noise?

In ideal cases, noise BW $\equiv$ Channel BW

$$N = BN_0 \quad \ldots (vi)$$

Energy per bit, $E_b = \frac{P}{R}$ ... (vii)

Or, $P = E_b R$ ... (viii)

From equation (iv), (vi) and (viii),

$$SNR = \frac{E_b R}{BN_0} = \frac{E_b}{N_0} \cdot \frac{R}{B}$$

$$= \frac{E_b}{N_0} \cdot \eta \quad \ldots (ix)$$

From equation (iii),

$$\eta = \log_2 (1 + SNR)$$

$$= \log_2 \left(1 + \frac{E_b}{N_0} \eta\right) \quad \ldots (x)$$

Or, $2^\eta = \left(1 + \frac{E_b}{N_0} \eta\right)$

On $\boxed{\frac{E_b}{N_0} = \frac{1}{\eta}\left(2^{\eta} - 1\right)}$

The ultimate relationship between cost of power efficiency and Bandwidth efficiency.

* $\eta \uparrow \rightarrow \frac{E_b}{N_0} \uparrow \rightarrow PE \downarrow \rightarrow B \downarrow$

* $\eta \downarrow \rightarrow \frac{E_b}{N_0} \downarrow \rightarrow PE \uparrow \rightarrow B \uparrow$

What happens to PE ? if $B \rightarrow \infty$ then $\eta \rightarrow 0$.

$$\lim_{\eta \to 0} \frac{E_b}{N_0} = \lim_{\eta \to 0} \frac{2^{\eta} - 1}{\eta}$$

$$\frac{\frac{d}{d\eta}\left(2^{\eta} - 1\right)}{\frac{d}{d\eta}(\eta)} = \frac{2^{\eta} \cdot \ln 2 - 0}{1} = 2^{\eta} \ln 2$$

$\eta = 0$, $\frac{2^0 \cdot \ln 2}{1} = 0.693$ (cost of PE ↓ least)

* For highest PE, cost will be 0.693.

$10 \log_{10}(0.693) = -1.59\ dB$

~~Q) PE~~ Q) Derive the least cost of Power efficiency

Q) What happens to PE if $B \rightarrow \infty$.

27-04-2026

Trade-off —

Hamming Code —

↳ Channel coding technique

↳ How many errors it can correct

↳ Linear block coding technique

↳ divide the message into blocks and then the blocks are passed

↳ It can correct and detect more than 1 bit error.

↳ Data word : actual data, no redundant bits (m)

Redundant bits : r

Code word = data word + redundant bits

$$2^r \geq m + r + 1$$

if $m = 7$ and $r = 2$ — ✗

if $m = 7$ and $r = 4$ — ✓

↳ Even Parity and ODD Parity

↳ number of one's = even

↳ number of one's = odd

* Data word = 1011001

$m = 7$, $r = 4$

P_1	P_2	D_1	P_4	D_2	D_3	D_4	P_8	D_5	D_6	D_7
1	0	1	0	0	1	1	1	0	0	1
1	2	3	4	5	6	7	8	9	10	11
2^0	2^1		2^2				2^3			

2^i *; $i \geq 0$

1. 0001
2. 0010
3. 0011
4. 0100
5. 0101
6. 0110
7. 0111
8. 1000
9. 1001
10. 1010
11. 1011

$P_1 = (1, 3, 5, 7, 9, 11)$ (least significant bit = 1)

1 1 0 1 0 1 ~~Even~~

$P_2 = (2, 3, 6, 7, 10, 11)$ ~~ODD~~ (ODD = 1, EVEN = 0)

0 1 1 1 0 1

$P_4 = (4, 5, 6, 7)$

0 0 1 1 ~~ODD~~

$P_8 = (8, 9, 10, 11)$ ~~Even~~

1 0 0 1

1 0 1 0 0 1 1 1 0 0 0

$\bar{S}_1$ $\bar{S}_2$ $\bar{S}_4$ $\bar{S}_8$

$S_1 = (1, 3, 5, 7, 9, 11) = 1$ (even)

~~0~~ 1 0 1 0 0

1

$S_2 = (2, 3, 6, 7, 10, 11) = 1$

1 1 1 0 0

$S_4 = (4, 5, 6, 7) = 0$

0 0 1 1

$S_8 = (8, 9, 10, 11) = 1$ } Syndrome Bits

1 0 0 0

~~$S_1 S_2 S_4 S_8$~~

1 1 0 1 = 11 (Decimal)

$S_8 S_4 S_2 S_1$ error position

1 0 1 1

1 error — Hamming Code

Multiple error — take another redundant bit

P_0 SECDED

S_8	S_4	S_2	S_1	P_0	
0	0	0	0	Correct	(No error)
≠ 0	0	0	0	Incorrect	(1 bit error)
≠ 0	0	0	0	Correct	(>1 bit error)
0	0	0	0	Incorrect	(Error in parity)

CRC —

↳ Take datawond

↳ Add redundant bits

↳ divide with particular value

$$n) a+b \; (\quad \cdots \quad b$$

$a+b-n \rightarrow$ if connect → divisible by n

if not → error

↳ drawbacks : We can detect multiple errors but can't connect it.

Polynomial :

$$g(x) = a_1 x^n + a_2 x^{n-1} + \ldots + a_n$$

$$a_1 \neq 0$$

$$g(x) = x^4 + x^3 + x^2 + x^1 + 1$$

(coefficient $0 \rightarrow 0$, $n \neq 0 \rightarrow 1$)

$x^4 + x^3 + x^2 + \quad 1$

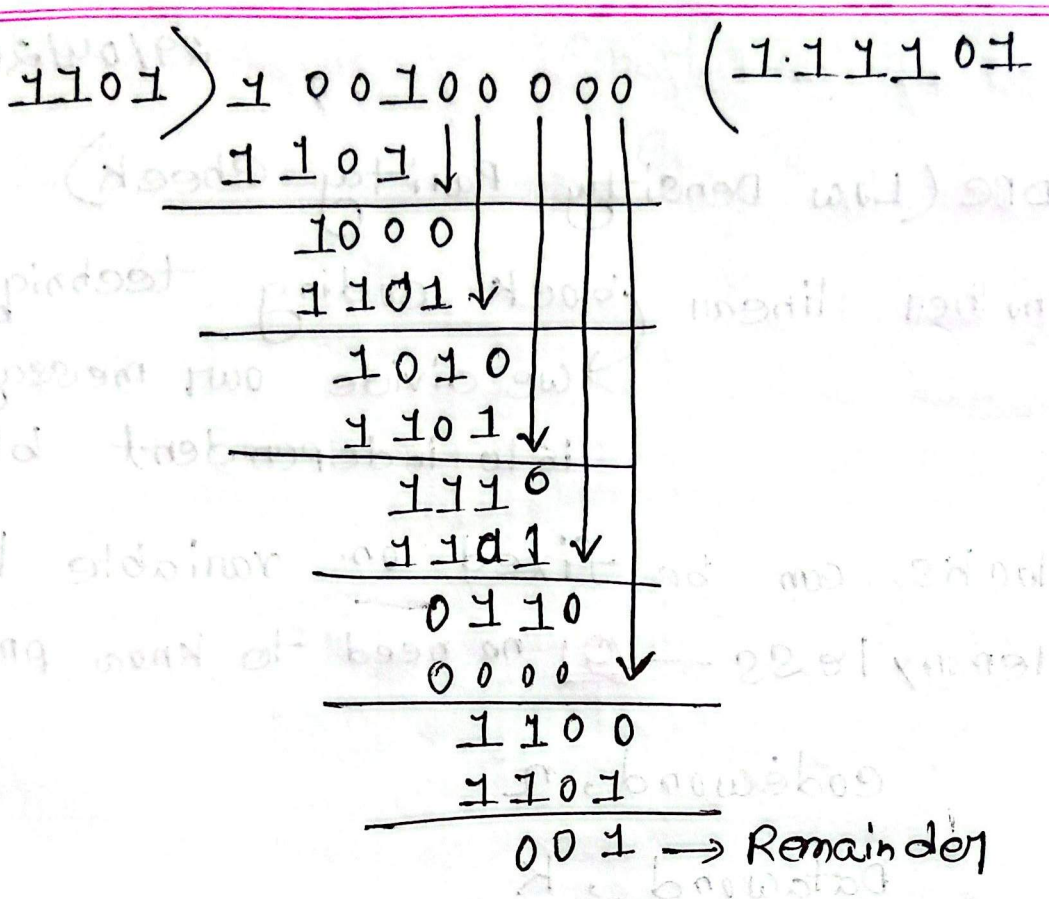
$g(x) = 1 \quad 1 \quad 1 \quad 0 \quad 1$

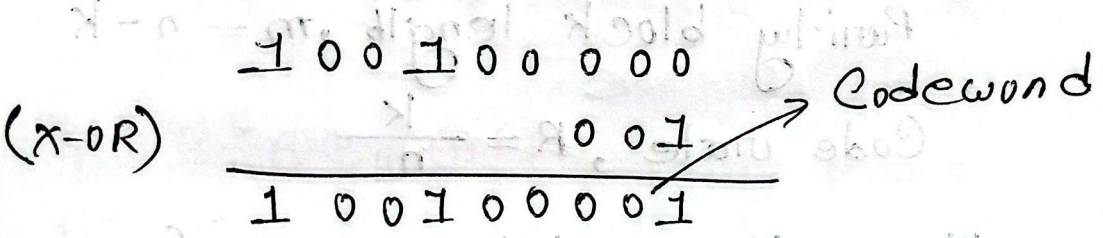
1 0 0 1 0 0 → data wond

add 0 (number = highest power)

if $g(x) = x^3 + x^2 + 1$

1 1 0 1


1101) 1 0 0 1 0 0 0 0 0
1 1 0 1
1 0 0 0
1 1 0 1
1 0 1 0
1 1 0 1
1 1 1 0
1 1 0 1
0 1 1 0
0 0 0 0
1 1 0 0
1 1 0 1
0 0 1 → Remainder
(1 1 1 1 0 1


1 0 0 1 0 0 0 0 0
(X-OR) 0 0 1
1 0 0 1 0 0 0 0 1
Codeword

29/04/2026

LDPC (Low Density Parity Check)

→ Another linear block coding technique
  ↳ we divide our message into independent blocks

→ Blocks can be fixed on variable length

→ Memoryless — ∴ no need to know previous one

codeword, n

Dataword, k

Parity block length, $m = n - k$

Code rate, $R = \frac{k}{n}$

* No parity needed — error-free environment
  parity needed — Noisy channel

Example: $(7, 4) \rightarrow (k-i) \times n$
(n, k)

→ Parity Check Matrix — Sparse (H)
— both for transmitter and receiver

(i) Random Construction (Gallager Method)
(ii) Quasi-Cyclic Convention
(iii) Progressive Edge-growth (PEG)

(iv) Finite Geometry Construction

Example

$$H = \left[\begin{array}{cccc|ccc} c_1 & c_2 & c_3 & c_4 & c_5 & c_6 & c_7 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

c_1, c_2, c_3, c_4: data — Set bits randomly (Random Construction)

c_5, c_6, c_7: Parity — Identity Matrix

⇒ Encoding :

data word	Parity
1 0 1 1	□ □ □

$c_1 \oplus c_2 \oplus c_4 \oplus P_1 = 0$
$1 \quad 0 \quad 1$ ⟹ $P_1 = 0$

$c_1 \oplus c_3 \oplus c_4 \oplus P_2 = 0$
$1 \quad 1 \quad 1$ ⟹ $P_2 = 1$

$c_2 \oplus c_3 \oplus c_4 \oplus P_3 = 0$
$0 \quad 1 \quad 1$ ⟹ $P_3 = 0$

Codeword, $S = 1\,0\,1\,1\,0\,1\,0$

⇒ Decoding : $H \cdot S^T$ (Transpose of Code)

$$= \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0\oplus1\oplus1\oplus0\oplus1\oplus1\oplus0 \\ 0\oplus0\oplus0\oplus0\oplus0\oplus0\oplus0 \\ 1\oplus1\oplus0\oplus0\oplus0\oplus1\oplus1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$ Syndrome

~~$0 \oplus 1 \oplus 1 =$~~

if every entry in Syndrome is 0 → No error

If we flip 1st & bit of codeword —

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

→ Have Error

↳ 3rd row → Connect

⇒ Voting : How many wrong rows each bit appears

$C_1 \rightarrow \boxed{2}$ $C_4 \rightarrow \boxed{2}$

$C_2 \rightarrow 1$ ~~$C_5 \rightarrow$~~

$C_3 \rightarrow 1$ Consider max value and Flip

→ Iterative procedure

04/05/2026

LDPC Encoding :

1011,010, (7,4)

LDPC Decoding :

1011

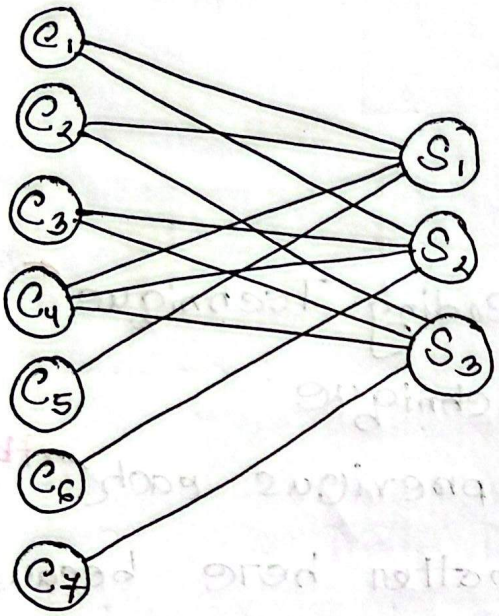
Error connection

$$HC^T = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

D_1

Voting is time consuming so we use Tanner Graph :



This graph is called 'Bipartite Graph'

Q) Why is Tanner Graph called Bipartite Graph?

Q) What is sparse matrix?

Q) Why H is called sparse matrix?

Q) Codeword: 1011 010 0111

Find parity bit for 2nd data word

Solve 0 0 1

$P_1 \oplus D_1 \oplus D_2 \oplus D_4 = 0$

$P_2 \oplus D_1 \oplus D_3 \oplus D_4 = 0$

$P_3 \oplus D_2 \oplus D_3 \oplus D_4 = 0$

Extended CRC :

* Drawback of CRC

Q) Most crucial drawback of CRC.

↳ CRC-32.
32 order
33-bit divisor

06/05/2026

Evaluation:

Hamming code
↳ Turbo code
↳ LDPC code
↳ Polar code (5G)

Polar Code — Block coding Technique — why?

→ most optimal technique
→ different from previous each — How?
→ Every bits matter here because contains information but it is overlapping information (redundancy)
→ Sequential decoding (No parallelism)
→ widely used in 5G
→ In Hamming, LDPC, parity bits position were fixed but not in Polar code.

Code rate, $R = \frac{K}{N}$

$N = 2^n$; $n = 1, 2, 4, 8, 16$

From the length of the codeword, we will

determine the length of the datawond.

if $R = 0.5$, $N = 4$ $\Rightarrow k = 0.5 \times 4 = 2$

In Hamming, CRC.... it was reversed

Data word, u

Code word, $x = u \cdot G_N$

G_N: Recursively defined matrix / Recursive Generator matrix

$$G_N = F^{\otimes n}$$

Kronecker Product

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$A \otimes F = \begin{bmatrix} aF & bF \\ cF & dF \end{bmatrix}$$

Case 1: $N = 2 = 2^1$

Then $G_N = F^{\otimes 1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Case 2: $N = 4 = 2^2$

Then $G_N = F^{\otimes 2} = F \otimes F = \begin{bmatrix} 1F & 0F \\ 1F & 1F \end{bmatrix}$

$$= \begin{bmatrix} F & 0 \\ F & F \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$N = 8 \rightarrow G_N = F \otimes F \otimes F$

$N = 8 = 2^3$

$G_N = G_8 = F^{\otimes 3} = F \otimes F \otimes F$

$= G_4 \otimes F$

$F^{\otimes n}$

$A^{\otimes N}$

$$G_1 = [1]$$

$$G_2 = \begin{bmatrix} G_1 & 0 \\ G_1 & G_1 \end{bmatrix} \quad G_4 = \begin{bmatrix} G_2 & 0 \\ G_2 & G_2 \end{bmatrix} \quad G_8 = \begin{bmatrix} G_4 & 0 \\ G_4 & G_4 \end{bmatrix}$$

* Q) The length of the codeword will always be the power of 2. Why?

$u = (0, 1)$ — 2 bits

$R = 0.5$

$N = ? = 4$

$2^n = 2^2 = 4$

$u_1 \; u_2 \; u_3 \; u_4$

$u_1 = 0$
$u_2 = 0$
$u_3 = 0$
$u_4 = 1$
} initially

$x = u \cdot G_4$

$= [u_1 \; u_2 \; u_3 \; u_4]$

$= [0 \; 0 \; 0 \; 1]$

$$\cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$x_1 = 0 \oplus 0 \oplus 0 \oplus 1 = 1$

$x_2 = 0 \oplus 1 \oplus 0 \oplus 1 = 1$

$x_3 = 1$

$x_4 = 1$

$x = [1 \; 1 \; 1 \; 1]$

Here, we have introduce a change in the codeword — difference from others.

11/05/2026

$y = [1\ 0\ 0\ 0]$

$u_1 \rightarrow u_2 \rightarrow u_3 \rightarrow u_4$

u_2 depends on u_1

u_3 depends on u_2

u_1	u_2	u_3	u_4
0	0	0	1

0 0 0 0 0 0 1 0

$x_1 = u.G_4$ $x_2 = u.G_4$

$$x_1 = [0\ 0\ 0\ 0] \cdot \begin{bmatrix} 1\ 0 & 0\ 0 \\ 1\ 1 & 0\ 0 \\ 1\ 0 & 1\ 0 \\ 1\ 1 & 1\ 1 \end{bmatrix} = [0\ 0\ 0\ 0]$$

$$x_2 = [0\ 0\ 1\ 0] \cdot \begin{bmatrix} 1\ 0 & 0\ 0 \\ 1\ 1 & 0\ 0 \\ 1\ 0 & 1\ 0 \\ 1\ 1 & 1\ 1 \end{bmatrix} = [1\ 0\ 1\ 0]$$

Compare x_1 and y

$x_1 = 0\ 0\ 0\ 0$

$y = 1\ 0\ 0\ 0$

1

Compare x_2 and y

$x_2 = 1\ 0\ 1\ 0$

$y = 1\ 0\ 0\ 0$

1

(i) Choose the 1st valid option

(ii) Use additional bit checks

(iii) Use reliability order.

(iv) Use soft values.

Q) Determine $u_3, u_4 \ldots$ show mismatch.

$y = [+0.9, -0.2, -0.1, +1.3]$

CT-3

Q) A 7-bit data-word is 1011001

use extended Hamming code. Find the final transmitted codeword including P_0.

Sol. – $m = 7$, $n = 4$

with parity, codeword $= m + n + 1$

$= 7 + 4 + 1 = 12$

P_0	P_1										

CamScanner

$u = (1, 1)$

$u_1 = 0 \quad u_2 = 0 \quad u_3 = 1 \quad u_4 = 1$

$$\text{Final} = [u_1\ u_2\ u_3\ u_4] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= [0\ 1\ 0\ 1]$$

* Receiver → 0 0 0 1

$$x_1 = [0\ 0\ 0\ 0] \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} = [0\ 0\ 0\ 1]$$

$$x_2 = [0\ 0\ 1\ 0] \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= [1\ 1\ 1\ 0]$$

$x_1 = 0\ 0\ 0\ 1$ $\qquad$ $x_2 = 1\ 1\ 1\ 0$

$y = 0\ 0\ 0\ 1$ $\qquad$ $y = 0\ 0\ 0\ 1$

1 ~~No~~ mismatch $\qquad$ 3 mismatch

Q1 ~~No~~ Some mismatch হলেও value suggest করতে হবে সঠিক Math

Time & Space Complexity —

For Hamming code —

i) $2^m > m+n+1 \rightarrow O(\log_2 n) \rightarrow O(n)$

ii) $n \longrightarrow O(n \times k)$

iii) n (Syndrome) $\rightarrow O(n \times k)$

Time Complexity $= O(n \times k)$

$= O(\log_2 n \times k)$

$\therefore$ Hamming Code $= O(k \log_2 n)$

For CRC —

Enc:

i) $m \times 1000$, $g(x) = x^3$ $\longrightarrow O(1)$

ii) $O(k \times m)$

iii) $O(k \times m$

$CRC = O(m \times k)$

$= O(k \log_2 n)$

For LDPC —

i) $O(n^3)$

ii) $(n, k) \rightarrow (k-1) \times n$

$O(n \times k-1 \times n)$

For Polar code —

G_4 $4 \times 4 \times 4$, $n \times n$ n $\rightarrow O(n^3)$

Q1 Compare the time complexity of the block coding techniques.

For space complexity — length of the codeword

18-05-2026

Hamming Code—

Encoding → $2^n \geq n+m+1$

$O(n \times n)$

Decoding → $O(n \times n)$

$O(n \times n)$

$2^n \geq n+m+1$

On, $2^n \geq n+1$

On, $2^n \approx n$

On, $\log_2 2^n \approx \log_2 n$

$n = \log_2 n$

$\therefore O(n \log_2 n)$

CRC — $O(kn)$

$g(x) = x^3 > m$

4 → 1011

$k \times m + 1$

1011 | 100100 | 000

LDPC — (n, k)

$O((n-k) \times n)$ → Parity bit

$O(n \times (k-1) \times n)$ → Syndrome

$O(n^3)$

$O(n^{2.3})$

} Encoding

$O(n \times (k-1) \times n)$ → Syndrome } Decoding

LDPC → $O(n \times (k-1) \times n)$

↳ $O(n^3)$

↳ $O(n^{2.3})$

Polar Code — $x = uG_N$

$O(N \times N \times N)$

↳ $O(N^3)$

↳ $O(N^{2.3})$

} Encoding

$O(N^3) \rightarrow O(N^{2.3})$ } Decoding

"Digital Modulation"

→ easier than analog modulation
need explicit demodulation

→ digital data to analog signal

→ Why need DM. — Long distance transmission

— Efficient Antenna size

$L \propto \lambda$ (wavelength)

↓

Antenna Height

$\lambda = \frac{c}{f}$

Antenna length will decrease if frequency of signal is increased.

Q1 From the perspective of Antenna Height, Explain the necessity of Digital Modulation

— Noise Immunity
— Multiplexing capability
— Efficient Bandwidth usage
— Compatibility with wireless communication

Digital Binary Modulation —

- changing 1 parameter (0/1) of carrier
- BASK, BFSK, BPSK → types
- Each bit → complete waveform (symbol)

* BASK/ASK —

1 — Carrier transmitted
0 — absent/lower

$$c(t) = A\cos(\underbrace{2\pi f_c t}_{\text{Angular variation}} + \underbrace{\phi}_{\text{Phase}})$$

Mathematical Representation:

$$s(t) = \begin{cases} A\cos(2\pi f_c t), & \text{for } 1 \\ 0, & \text{for } 0 \end{cases}$$

ϕ is truncated

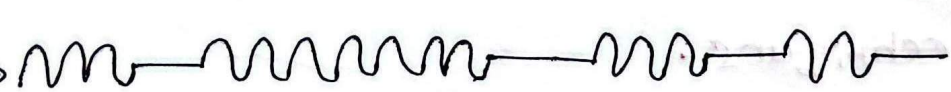
1 0 1 1 0 1 0 1



Carrier →



ASK →

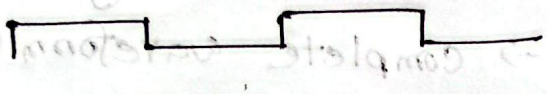


* BFSK / FSK —

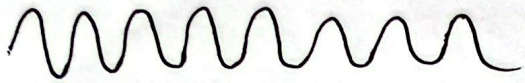
1 — f_1
0 — f_2

$$s(t) = \begin{cases} A\cos(2\pi f_1 t), & \text{for } 1 \\ A\cos(2\pi f_2 t), & \text{for } 0 \end{cases}$$

Msg –



Carrier –



FSK –



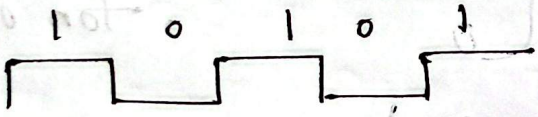
Q1 What are Pros, Cons of FSK?

* BPSK/PSK –

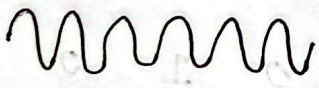
$$s(t) = \begin{cases} A\cos(2\pi f_c t), & \text{for } 1 \\ -A\cos(2\pi f_c t), & \text{for } 0 \end{cases} \Big) 180^\circ$$

1 0 1 0 1

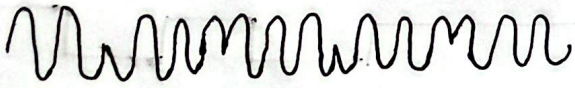
Msg –



Carrier –



PSK –

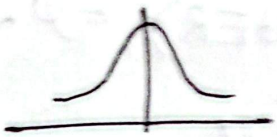


Q1 Compare all three binary modulation techniques.

AWGN

(i) Additive — $Y = X + N$

(ii) White — same power over freq. spectr

(iii) Gaussion —  Normal distrib

Source — Thermal property

Use —

(i) Baseline Performance — QAM, PSK

(ii) Building Block — Rayleigh, Rician — Noise

(iii) Simulation

AWGN	Real life
i) Thermal / Electronic	i) Interfe, Multipath
ii) Perfectly flat	ii) freq.
iii) low-fidelity	iii) Hy h (s-
iv) Benchmarks (initial design)	iv) Dy

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Coherent and Non-Coherent Receiver —

Receiver two types — i) Coherent
ii) Non-Coherent

↳ It affects —

i) Bit Error Rate (BER) $= Q\sqrt{\frac{2E_b}{N_b}}$

ii) Receiver Complexity

iii) Synchronization Requirements

iv) Power

Main Components of C.R. —

i) RF Front End (Antenna + Bandpass Filter)

* ii) Local Oscillator (Carrier Generator)
$\cos(2\pi f_c t + \phi)$

* iii) Mixer (Product Detector)

iv) Low Pass Filter (LPF)

v) Sampler / Integrator

vi) Decision Device

** Example

Non-Coherent Receiver —

— without requiring carrier phase synchronization
— No need of extra reference carrier
— Simpler, Cheaper, Easier to implement, poor BER

↳ Main Components :

i) Bandpass Filter
ii) Envelope detector / Energy Detector
iii) Integrator / Averager
iv) Decision Device

CamScanner